In the Mix - Developing open source search technologies on the Microsoft platform

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UK Data Service

Overview

- Why use open source?
- The Solr project
- Integrating Solr with Windows
- Application architecture
- Server architecture
- Debugging and monitoring
- Questions

Why use open source?

- Features
- Performance
- Community
- Roadmap
- Cost
- Tooling

Unresolved: By Assignee

Alan Woodward	2	
Andrzej Bialecki	2	
Bill Au	2	
Chris Male	1	
David Smiley	2	
Dawid Weiss	1	
Despot Jakimovski	1	
Erick Erickson	8	1%
Erik Hatcher	11	1%
Grant Ingersoll	8	1%

...and 22 more

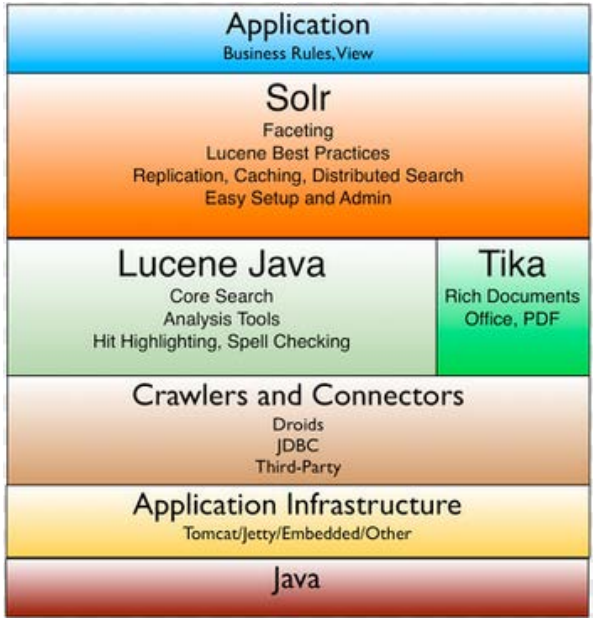
Unresolved: By Version

- 1.4.2
- 3.1.1
- 3.3
- 3.4
- 3.5
- 3.6
- 4.0-BETA
- 4.0
- 3.6.3
- 4.1

...and 6 more

The Solr project

- Open-source solution, largely supported by the Apache Software Foundation.
- Written in Java
- Created in 2004, current version 4.3
- SOLR works with Hadoop, Lucece, Tika, Nutch
- Capable of indexing billions of items in a clustered environment.
- Features include:
  - Full-text search
  - Faceted search
  - Highlighting
  - Rich document handling
  - Distributed search (Solr Cloud)
  - Highly scalable
  - NoSQL



Apache

Solr on Windows

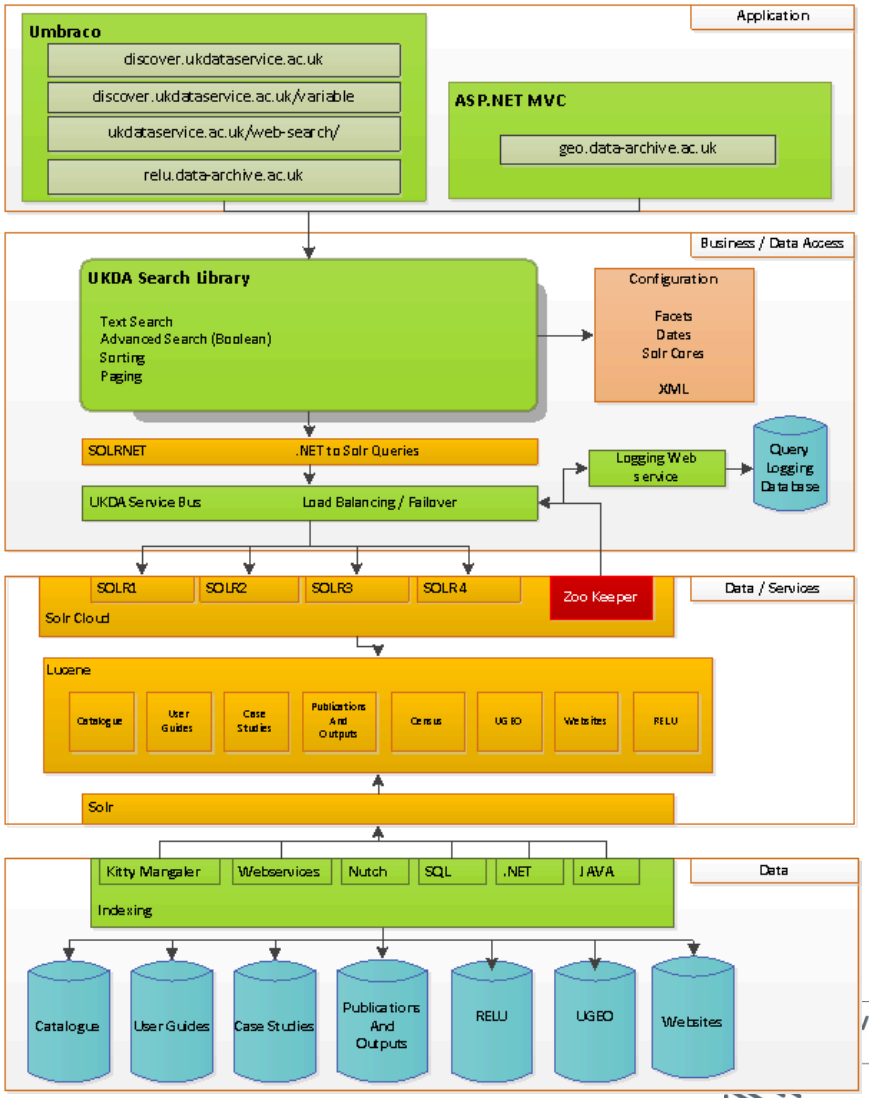
- Requires Java and web application server (Tomcat, Jetty)
- Rest web service API

  http://dasolr/solr/Catalogue/select? q=“Social Science"& sort=Date+desc

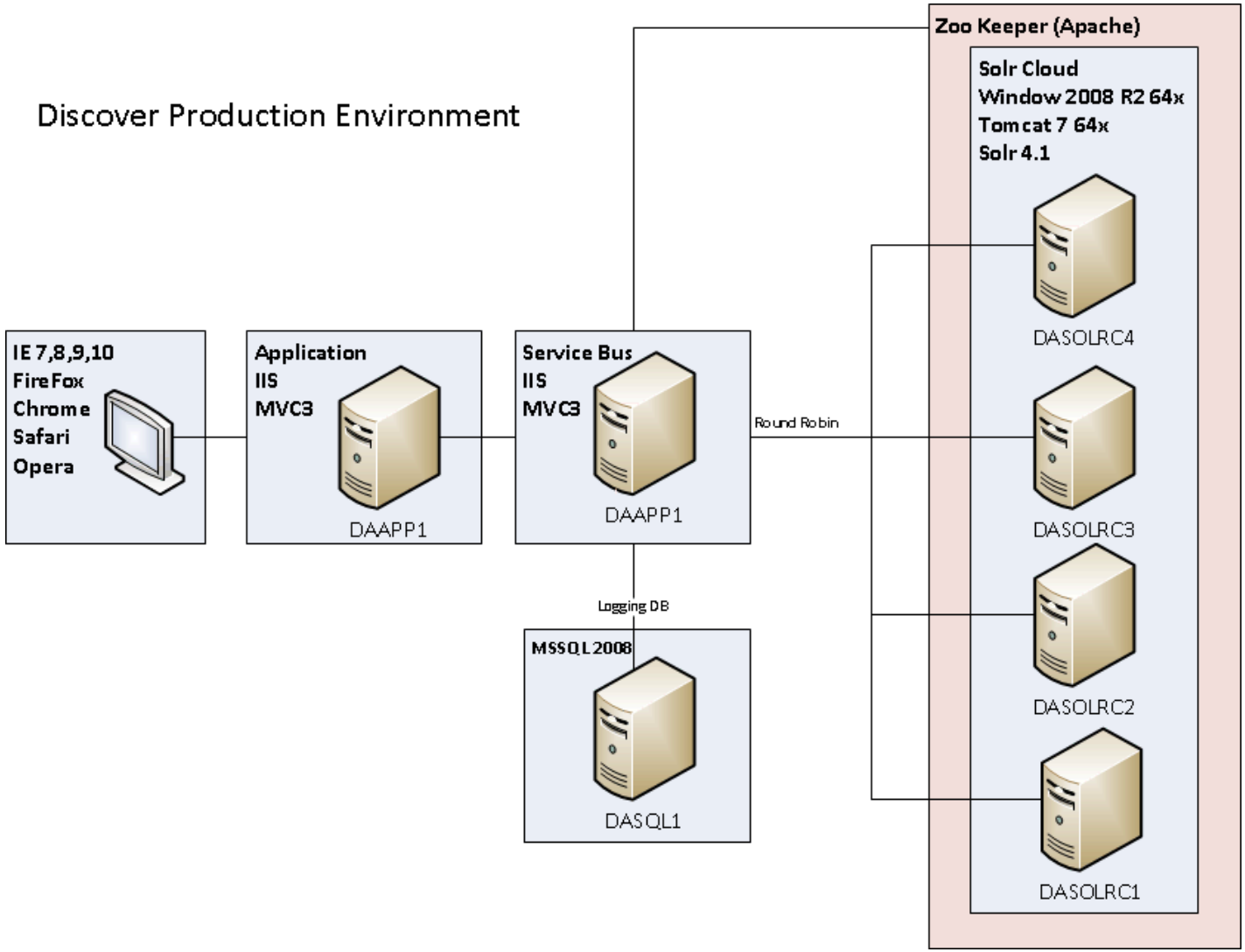
- Manual installation
- Plugins might only run on Linux or require an emulator (Cygwin)
- Configured through XML
- Reduce dependencies though Service Bus

Application Architecture

- **Application**
  - Umbraco / ASP.MVC
  - HTML
  - Jquery
- **Business / Data Access**
  - .NET Libraries
    - UKDA.Search.Library
    - SolrNet
    - Mule
- **Solr Cloud**
  - Solr 4.1
  - Lucene
  - JVM 7
  - Tomcat 7
- **Data**
  - MS SQL
  - .NET 4 Console
  - Entity framework / WebAPI



Server architecture



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Debugging and monitoring

- Query Logs (Enterprise Service Bus)
- Solr Logs
- Elmah (.NET error logging)
- Fiddler (HTTP Monitoring)
- WireShark (Network Monitoring)

Questions